Engineering Mathematics III - Fourier & Laplace Transforms

Department: Engineering Mathematics | Semester: 3 | Code: MA301
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1. Fourier Series & Dirichlet Conditions

Any periodic function f(x) satisfying Dirichlet conditions can be represented as: f(x) = a0/2 + sum(an*cos(nx) + bn*sin(nx)). Used to decompose complex signals into harmonic constituents.

2. Laplace Transforms & Initial Value Problems

Laplace transform converts differential equations in the time domain into algebraic equations in the s-domain: L{f(t)} = integral(0 to inf, e^(-st)*f(t) dt).

3. Partial Differential Equations (Wave & Heat)

One-dimensional wave equation (d^2 u/dt^2 = c^2 * d^2 u/dx^2) and Heat conduction equation solved using separation of variables.